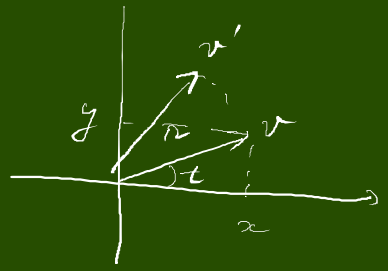
$$v = \begin{pmatrix} x \\ y \end{pmatrix} \longmapsto Av = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$= \begin{pmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{pmatrix}$$

$z = x + iy$

$z = r(\cos t + i \sin t) \longrightarrow z' = r(\cos(t+\theta) + i\sin(t+\theta))$



$= r(\cos t \cos\theta - \sin t \sin\theta) +$
$ir(\sin t \cos\theta + \sin\theta \cos t)$

$= x\cos\theta - y\sin\theta +$
$i(x\sin\theta + y\cos\theta)$

$f : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$

$f(v) = Av$ notação de ângulo

$v = \begin{pmatrix} x \\ y \end{pmatrix} \longmapsto \begin{pmatrix} x \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

temă $\quad y = \lambda x \quad A = ?$

$v = \begin{pmatrix} x \\ y \end{pmatrix} \longmapsto \begin{pmatrix} -x \\ y \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

$f(\lambda x + \mu y) = \lambda f(x) + \mu f(y) \quad \forall x, y \in U$

$f : \mathbb{R}^2 \longrightarrow \mathbb{R}^2 \quad f\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+1 \\ y-1 \end{pmatrix}$ nu e aplic. lin.

$f\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x^3 \\ y \end{pmatrix}$

Exemple. 1) $f : \mathbb{R}[X] \longrightarrow \mathbb{R}$

$f(P) = P(3)$ aplic. lin?

$\mathcal{J}(P+Q) = (P+Q)(3) = P(3)+Q(3) = \mathcal{J}(P)+\mathcal{J}(Q)$

$\mathcal{J}(\lambda P) = (\lambda P)(3) = \lambda P(3) = \lambda \mathcal{J}(P)$

2) $\mathcal{J} : \mathbb{R}[x] \longrightarrow \mathbb{R}[x]$

$\mathcal{J}(P) = P'$

$P(x) = a_0 + a_1 x + \dots + a_n x^n$

$\mathcal{J}(P)(x) = P'(x) = a_1 + 2a_2 x + \dots + n a_n x^{n-1}$

$\mathcal{J}(P+Q) = \mathcal{J}(P)+\mathcal{J}(Q)$

$(P+Q)' = P'+Q'$ ✓

$\mathcal{J}(\lambda P) = \lambda \mathcal{J}(P)$

$(\lambda P)' = \lambda P'$ ✓

3) $\mathcal{J} : U \longrightarrow V$ $\quad \mathcal{J}(\varphi) = \varphi'$ aplic. liniară

$U = \{ \varphi : \mathbb{R} \longrightarrow \mathbb{R} \mid \varphi \text{ derivabilă} \}$

$V = \{ \psi : \mathbb{R} \longrightarrow \mathbb{R} \mid \psi \text{ funcție} \}$

4) $g : \mathbb{R}[X] \longrightarrow \mathbb{R}[X]$

$$g(P) = \int_0^X P(t)\,dt$$

$$P(X) = a_0 + a_1 X + \dots + a_n X^n$$

$$g(P)(X) = a_0 X + \frac{a_1}{2} X^2 + \dots + \frac{a_n}{n+1} X^{n+1}$$

$f \circ g = id$? $\quad g \circ f = id$?

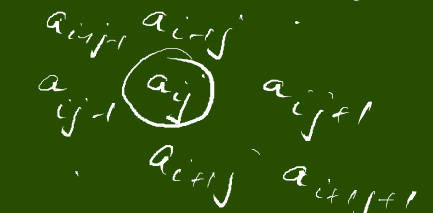
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5) $f : \mathcal{M}_{m,n}(\mathbb{R}) \longrightarrow \mathcal{M}_{m,n}(\mathbb{R})$

$$f(A)_{ij} = \frac{1}{9}\left(a_{i-1\,j-1} + \dots + a_{i+1\,j+1}\right)$$

$A = (a_{ij})_{\substack{i=\overline{1,m} \\ j=\overline{1,n}}}$

$A + B$



$$f(A+B) = f(A) + f(B)$$
$$f(\lambda A) = \lambda f(A)$$

$\mathbb{1}_V$ $\quad$ Id_V $\quad$ id_V

$$V \longrightarrow V$$
$$v \longmapsto v$$
$$v \longmapsto 0$$

---

f bijective/date $\longrightarrow$ f^{-1} bij

f prop. $\longrightarrow$ f^{-1} prop. ?

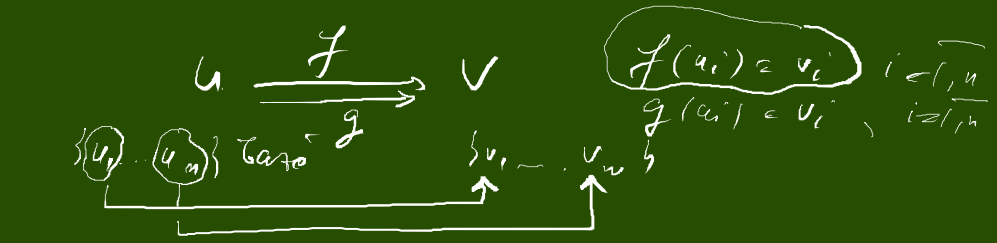
cont $\qquad$ cont ?

- $f : \mathbb{R} \longrightarrow \mathbb{R}$ cont, bij $\overset{\Delta}{\Longrightarrow} f^{-1}$ cont!
- $f : A \subseteq \mathbb{R} \longrightarrow B \subseteq \mathbb{R}$ $\overset{\Delta\Delta}{\Longrightarrow} f^{-1}$ cont!

$$f : [0,1] \cup (2,3] \longrightarrow [0,2]$$

$$f(x) = \begin{cases} x, & x \in [0,1] \\ x-1, & x \in (2,3] \end{cases}$$

$$f^{-1}(y) = \begin{cases} y, & y \in [0,1] \\ y+1, & y \in (1,2] \end{cases}$$



$f(u_i) = v_i \quad i \in \overline{1,n}$

$g(v_i) = u_i, \quad i \in \overline{1,n}$

mincitate

$$f(u) = g(u) \quad \forall \; u \in U$$

Fie $u \in U$

$\mathcal{B} = \{u_1, \dots u_m\}$ baza $\Rightarrow \exists \lambda_1 \dots \lambda_m \in K$ $u = \lambda_1 u_1 + \dots + \lambda_m u_m$

$$\Rightarrow f(u) = f(\lambda_1 u_1 + \dots + \lambda_m u_m) = \lambda_1 f(u_1) + \dots + \lambda_m f(u_m)$$

$= \lambda_1 g(u_1) + \dots + \lambda_n g(u_n) = g(\lambda_1 u_1 + \dots + \lambda_n u_n) = g(u)$

$\Rightarrow f = g$.

existență: fie $u \in U$ $\quad f(u) = ?$

$u = \lambda_1 u_1 + \dots + \lambda_n u_n$

$f(u) = \lambda_1 v_1 + \dots + \lambda_n v_n$

$[u+u']_B = [u]_B + [u']_B$

$\downarrow$

$f(u+u') = f(u) + f(u')$

$[\lambda u]_B = \lambda [u]_B$

$\downarrow$

$f(\lambda u) = \lambda f(u)$

---

f injectivă $\Leftrightarrow$ $f(u_1), \dots, f(u_n)$ liniar independente.

$U \xrightarrow{f} V$

$\boxed{f(u) = 0} \Rightarrow \boxed{u = 0}$

f injectivă $\Leftrightarrow$ $\ker(f) = \{0_U\}$

"$\Rightarrow$" Fie $u \in \ker(f) \Rightarrow f(u) = 0_V$

f aplicație liniară $\Rightarrow f(0_U) = 0_V$ $\Rightarrow$

f injectivă

$\Rightarrow u = 0_U \Rightarrow \ker(f) = \{0_U\}$

"$\Leftarrow$" Fie $u, u' \in U$ a.î. $f(u) = f(u')$

$f(u) - f(u') = 0_V$

$f(u - u') = 0_V$

$u - u' \in \ker(f) = \{0_U\}$

$u - u' = 0_U$

$u = u' \Rightarrow f$ inj.

$A = \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 0 \end{pmatrix}$ $\quad \ker A \subseteq \mathbb{R}^n$

$\ker A = \mathbb{R}^n$

$A = (a_{ij})_{\substack{i=\overline{1,m} \\ j=\overline{1,n}}}$ $\quad \ker(A) = \{0_n\}$

$\Leftrightarrow \operatorname{rang} A = n \leq m$

$\Leftrightarrow$ coloanele lui A sunt lin. indep.

$\operatorname{Im}(A) = \mathbb{K}^m \Leftrightarrow \operatorname{rang} A = m \leq n$

$Sp\{\text{col lui } A\} \Leftrightarrow$ Col. lui A Syst. de gen. pt $\mathbb{K}^m$

$k + \underline{?} = n$

$f : U \longrightarrow V$ aplic. lin.

$\ker(f) \subseteq U \qquad \operatorname{Im}(f) \subseteq V$

bază $\{u_1 \dots u_k\}$ în $\ker(f)$

completăm $\{u_1 \dots u_k, u_{k+1}, \dots, u_n\}$ bază în U

$\downarrow f$

~~$f(u_1) \dots f(u_k)\}$~~ $\{f(u_{k+1}), \dots, f(u_n)\}$ bază

$\in \mathrm{Im}(f)$

liniar independ. | sist. gener.

Exemplu: $f : M_2(\mathbb{R}) \to M_2(\mathbb{R})$

$$f(A) = \frac{A + A^t}{2}$$

a) ex: f aplic. liniară.

b) $\ker f = ?$ $\mathrm{Im} f = ?$

o bază, dim.

c) $\dim \ker f + \dim \mathrm{Im} f = \dim_{\mathbb{R}} M_2(\mathbb{R}) = 4$

---

b) $\ker(f) = \{A \in M_2(\mathbb{R}) \mid f(A) = O_2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\}$

$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ $\quad f(A) = \frac{A + A^t}{2} = \begin{pmatrix} a & \frac{b+c}{2} \\ \frac{b+c}{2} & d \end{pmatrix}$

$f(A) = 0_2 \iff a = d = 0$
$b = -c$

$\ker(f) = \left\{ \begin{pmatrix} 0 & b \\ -b & 0 \end{pmatrix} / b \in \mathbb{R} \right\}$

matrice antisymétrique
$A^t = -A$

$= Sp\left\{ \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\}$

$\left\{ \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\}$ base de $\ker(f)$

$\dim_{\mathbb{R}} \ker(f) = 1$

$\mathrm{Im}(f) = \{ B \in \mathcal{M}_2(\mathbb{R}) / \exists A \in \mathcal{M}_2(\mathbb{R}), f(A) = B \}$

$B = \begin{pmatrix} x & y \\ z & t \end{pmatrix}$ $\quad f(A) = B \quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$\Rightarrow \begin{cases} a = x ; d = t \\ \frac{b+c}{2} = y \\ \frac{b+c}{2} = z \end{cases}$

$\Rightarrow \dim = 3$

$\operatorname{Im}(f) = \{ \underbrace{\begin{pmatrix} x & y \\ y & t \end{pmatrix}}_{\text{matrice le simetrice}} \mid x, y, t \in \mathbb{R} \}$

$B^t = B$

$\begin{pmatrix} x & y \\ y & t \end{pmatrix} = x \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + y \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + t \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

$\operatorname{Im}(f) = \operatorname{Sp} \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$

$\Rightarrow$ bază în $\operatorname{Im}(f)$ $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$

$\Rightarrow \dim_{\mathbb{R}} \operatorname{Im}(f) = 3$

$\Rightarrow \underbrace{\dim_{\mathbb{R}} \ker(f)}_{1} + \underbrace{\dim_{\mathbb{R}} \operatorname{Im}(f)}_{3} = \underbrace{\dim_{\mathbb{R}} M_2(\mathbb{R})}_{4}$ ✓

Exercițiu :— $f : \underbrace{\mathbb{R}_2[X]}_{\dim_{\mathbb{R}} \mathbb{R}_2[X] = 3} \longrightarrow M_2(\mathbb{R})$

$$\mathcal{J}(P) = \begin{pmatrix} P(1) & P'(1) \\ 0 & P(1)+P'(1) \end{pmatrix}$$

a) $\mathcal{J}$ aplic. liniară

b) $\ker(\mathcal{J}) = ?$ $\operatorname{Im}(\mathcal{J}) = ?$

(bază, dim.

⇒ inj. ?

$\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$ ⇄ $P(X) = a + bX + cX^2 + dX^3$

$\mathbb{R}^4 = R_3[X] \simeq M_2(\mathbb{R})$

$e_1, e_2, e_3, e_4 \longleftrightarrow 1, X, X^2, X^3$

$E_{11}, E_{12}, E_{21}, E_{22}$

Thm. $U \simeq V \iff \dim_K U = \dim_K V$

$\exists\, f : U \to V$ izomorfism

Dem. $\Rightarrow$ "

$\dim_K \ker(f) + \dim_K \operatorname{Im}(f) = \dim_K U$

$\ker(f) = \{0\}$; $\operatorname{Im}(f) = V$

$0 + \dim_K V$

$\Leftarrow$ "u"

$\dim_k U = \dim_k V = n$

$\mathcal{B} = \{u_1 \dots u_n\}$ bază în U

$\mathcal{B}' = \{v_1 \dots v_n\}$ bază în V

$\Rightarrow \exists!\ f : U \longrightarrow V$ aplic. lin

$f(u_i) = v_i \quad i = \overline{1,n}$

$\exists!\ g : V \longrightarrow U$ aplic. lin.

$g(v_i) = u_i \quad i = \overline{1,n}$

$g \circ f : U \longrightarrow U \quad (g \circ f)(u_i) =$

aplic. lin $\quad U \xrightarrow{f} V \xrightarrow{g}$

$= g(f(u_i)) = g(v_i) = u_i$

$(g \circ f)(u_i) = u_i \quad i = \overline{1,n}$

$id_U : U \longrightarrow U$

$id_U(u_i) = u_i \quad i = \overline{1,n}$

$\Rightarrow g \circ f = id_U$

analog $f \circ g = id_V$

$\Rightarrow f$ invertierbar (bij.) $f^{-1} = g$.

$\Rightarrow f$ Isom.